

Chhavi Kirtani

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EDUCATION

University of California San Diego

MS in Computer Science - TA for Statistical NLP Course, RA under Prof Paturi

Coursework: AI Agents, Recommender systems, Computer Vision, Advances in NLP research, Convex Optimization

San Diego, California, United States

Sept 2025 – June 2027

IIIT Delhi

BS in Computer Science and Applied Mathematics with Honors (3.9/4 GPA)

Honors Degree-awarded for research and academic distinction (top 7% of graduates)

Delhi, India

Aug 2018 – June 2022

EXPERIENCE

AI/ML Researcher

Feb'24 - Mar'25 & Aug'20 - June'22

Under Dr. Rajiv Ratn Shah, Dr. Dhruv Kumar, Dr. Rajitha Prasad

Remote

- **AAAI 2025 Publication (Oral Presentation: 5% acceptance rate): Outperformed Gemini 1.0 Pro** in English and Hindi **Math Reasoning using only Small LLMs (SLM, 8B parameters)**. Introduced a **novel transfer learning paradigm**: bilingual curriculum fine-tuning + rehearsal based learning to mitigate catastrophic forgetting + structure decomposition for arithmetic reasoning.
- **EMNLP 2025 Findings**: Built a **multi-agentic LLM research reviewer with self-critiquing loop**, achieving **10–20% higher quality than AllenAI MARG & Sakana AI**; designed **Novel Hallucination & Insight Evaluation Metrics**
- Improved **multimodal Physics QnA accuracy of Small Vision Transformer models by 35%** by leveraging **representation learning using Infi-MM captioning** to generate **visual descriptions** for scientific diagrams, combined with **RLHF and reward modeling**.
- Built an end-to-end **Computer Vision pipeline for Brand and Logo Recognition**, achieving **94% accuracy** using **YOLOv5 object detection** and **OCR-based text recognition** on large-scale product imagery.
- **Led a team to build an Agentic AI chatbot for U.S. Discrimination Law**, designing a **RAG pipeline** with **Neo4j knowledge graphs** and **Pinecone** for legal retrieval and reasoning. **Fine-tuned LegalRoBERTa** on **5K+ scraped & custom domain Q&A pairs**. [\[GitHub\]](#)
- **Graph-Based Survival Analysis** (Bachelor's Thesis, A): Built a **causally-structured DAG-based deep survival model** combining **statistical survival analysis & neural networks** to model covariate dependencies, achieving **10% improvement** over SOTA baselines.
- **Tech Stack**: Python, PyTorch, TensorFlow, LORA, PEFT, Transformers, HuggingFace, LangChain, Neo4j, Pinecone, OpenCV, FastAPI

Senior Software Engineer

July'22 – Sept'25

NatWest Group (Fraud Prevention Department)

Gurugram, India

- Improved scalability, fault isolation, and CI/CD agility by **refactoring a 4K+ LOC fraud detection monolith software** into **6 distributed Spring Boot microservices** and host on **AWS**. Achieved **98% unit test coverage**, driving a **60% drop in production issues**.
- Boosted **fraud action throughput 3×** by redesigning an **asynchronous Kafka data pipeline using Apache Beam** and modernizing **streaming microservices** (Java 8→17); **led team-wide Splunk logging overhaul** to **eliminate blind spots** and **optimize monitoring**.
- Built a full-stack investment tracker end-to-end system (**Javascript, React, Java**) with portfolio management, real-time marketwatch, trading features, and **interactive UI**, earning **Top 5 for UX at NatWest Hackathon**. [\[UI Mockups\]](#) [\[GitHub\]](#)
- Reduced **database costs by 80%** with **<5 min downtime** by migrating **2 PCF microservices** from **Oracle to PostgreSQL**, refactoring **JDBC queries** for Postgres compatibility, and validating **1M+ records** with **Python+SQL ETL scripts**.
- Received **2 Loving Your Work Awards** for ownership and timely production releases. **Tech Stack**: Java, Spring Boot, AWS, Kafka, Apache Beam, PostgreSQL, Oracle, PCF, Python, SQL, React, JavaScript, Splunk, CI/CD

Software Engineer Intern

May'21 – July'21

Amazon (Performance Advertising Department)

Bangalore, India

- **Architected** and Deployed **2 Java Microservices** to **Support Advertisers on Amazon Marketplace** featuring **RESTful public-facing APIs**, implementing **paginated response handling** to ensure low-latency data retrieval for large-scale datasets.
- Built a **Python** data migration tool that accelerated **DynamoDB backfilling by 18x** using **multiprocessing** and batch writes.

Computer Vision Engineer (Open-Source Contributions)

May'24 - Dec'24

Omdena

Remote

- **Low-Cost 3D Reconstruction for Cultural Heritage**: Built a **multi-view graphics pipeline to reconstruct 3D models** of endangered artifacts from ordinary photos. Used **stereo geometry (RANSAC)** and illumination modeling to **recover surface normals and motion**.
- **Ghanaian Sign Language Translator**: Developed an **LSTM gesture classifier** on MediaPipe keypoints; designed geometric augmentation (perspective/viewpoint simulation) to **boost accuracy from 13% to 83%**. [\[GitHub\]](#)
- **Crop Yield Prediction (Kenya) using satellite imagery**: Built ML models using **8 years of agro-climatic satellite data (47 counties)** from **Google Earth Engine** to predict maize yield (94.5% accuracy); performed feature engineering for robust generalization.
- **Real-Time Hand-Gesture Controlled Game Interface (Hackathon Winner)**: Built a human-machine interaction system integrating a Python game with an **Arduino distance sensor for gesture-based control** via serial communication. [\[GitHub\]](#)

SELECTED PUBLICATIONS

SEE ALL PUBLICATIONS AT: CHHAVIKIRTANI2000.GITHUB.IO

- **Multilingual Mathematical Reasoning: Advancing Open-Source LLMs in Hindi and English**: [\[Published\]](#) in AAAI'25 (Main Track, Oral) Avinash Anand, Kritarth Prasad, Chhavi Kirtani, Ashwin R Nair, Manvendra Kumar Nema, et al;
- **ReviewEval: An Evaluation Framework for AI-Generated Reviews**: [\[Published\]](#) in EMNLP 2025 (Findings) Madhav Krishnan Garg, Tejash Prasad, Tanmay Singhal, Chhavi Kirtani, Murari Mandal, Dhruv Kumar;
- **MM-PhyRLHF: Reinforcement Learning Framework for Multimodal Physics Question-Answering** [\[Arxiv\]](#) Janak Kapuriya*, Chhavi Kirtani*, Apoorv Singh, Jay Saraf et al; * Equal Contribution
- **Population / patient phenotype similarity based GCN for statistical survival analysis** [\[Bachelor Thesis\]](#) Chhavi Kirtani, Rajitha Anand